Assignment 1 — Structured Data (CSV)

Create a file called:

```
library_loans.csv
```

Paste this data:

```
Loan_ID,Member_Name,Book_Category,Days_Borrowed,Fine
L001,Nora,Fiction,7,0
L002,Dev,Science,18,120
L003,Ibrahim,History,12,40
L004,Meera,Fiction,21,180
L005,Rohan,Technology,9,0
L006,Sana,Science,15,70
```

Create:

```
structured_assignment.py
```

Tasks

1. Load the CSV into a Pandas DataFrame.
2. Display only `Member_Name`, `Book_Category`, and `Fine`.
3. Show only members whose `Fine` is greater than 50.
4. Create a new column called `Status`:
    - `No Fine` if Fine = 0
    - `Fine Due` otherwise
5. Find the **average number of days borrowed**.
6. Find the **total fine collected by Book_Category**.

The goal is to practise:

```
CSV → DataFrame → Filter → New Column → Group
```

Assignment 2 — Semi-Structured Data (JSON)

Create:

```
festival_events.json
```

Paste:

```
[
  {
    "event_id": "E101",
    "name": "Food Carnival",
    "location": {
      "city": "Kochi",
      "venue": "Marine Ground"
    },
    "visitors": 850,
    "features": ["food", "music"]
  },
  {
    "event_id": "E102",
    "name": "Book Fair",
    "location": {
      "city": "Bhopal",
      "venue": "Central Hall"
    },
    "visitors": 620
  },
  {
    "event_id": "E103",
    "name": "Tech Expo",
    "location": {
      "city": "Ahmedabad",
      "venue": "Expo Centre"
    },
    "visitors": 1200,
    "features": ["AI", "Robotics", "Startups"]
  }
]
```

Create:

```
semi_structured_assignment.py
```

Tasks

1. Load the JSON using Python.
2. Print the name of every event.
3. Print the city of each event.
4. Display only events with more than 700 visitors.
5. Some events do not have `features`. Display:
    - the features if available
    - `"No features listed"` if missing
6. Convert the JSON into a Pandas DataFrame using `pd.json_normalize()`.

The goal is to understand:

```
JSON
 ↓
List / Dictionary
 ↓
Nested Fields
 ↓
Optional Fields
 ↓
DataFrame
```

Assignment 3 — Unstructured Data (Text)

Create:

```
wildlife_report.txt
```

Paste:

```
Forest officers observed a herd of elephants near the eastern water point

Two deer were seen close to the grassland area.

A fallen tree blocked part of the northern trail.
```

```
The patrol team also noticed fresh elephant footprints near the river.

No injured animals were reported.

The northern trail will remain closed until the fallen tree is removed.
```

Create:

```
unstructured_assignment.py
```

Tasks

1. Read the complete text file.
2. Count the total number of words.
3. Check whether the word `elephant` appears in the report.
4. Count how many times `elephant` appears.
5. Print only the lines containing the word `trail` .
6. Print only the lines containing the word `animal` .

The goal is to understand:

```
Free-form text
      ↓
Read
      ↓
Search
      ↓
Count
      ↓
Extract useful information
```